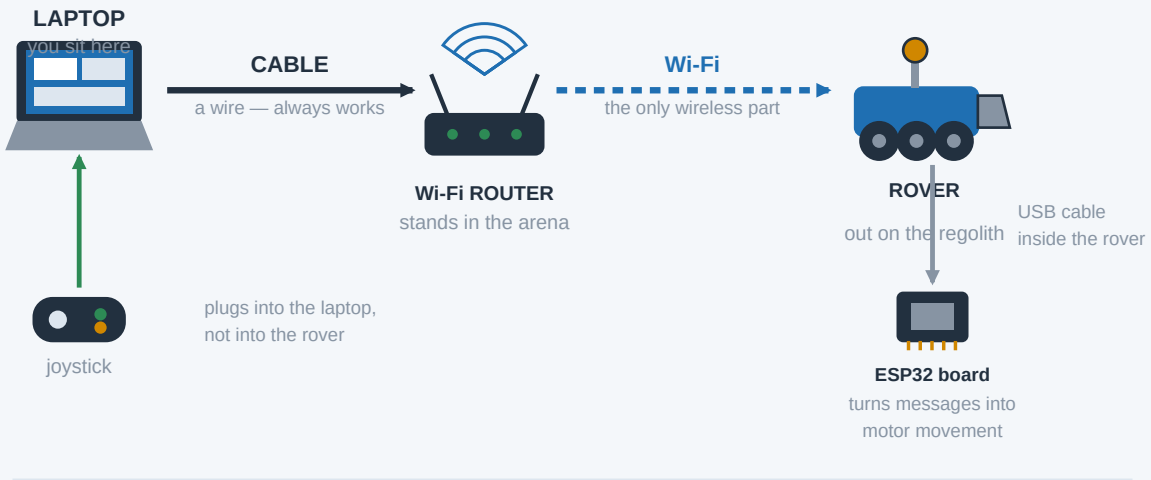
# How the laptop talks to the rover

A picture-first explanation · October 2026




**One wire, one radio hop, one USB cable. That is the whole path.**

## The short version

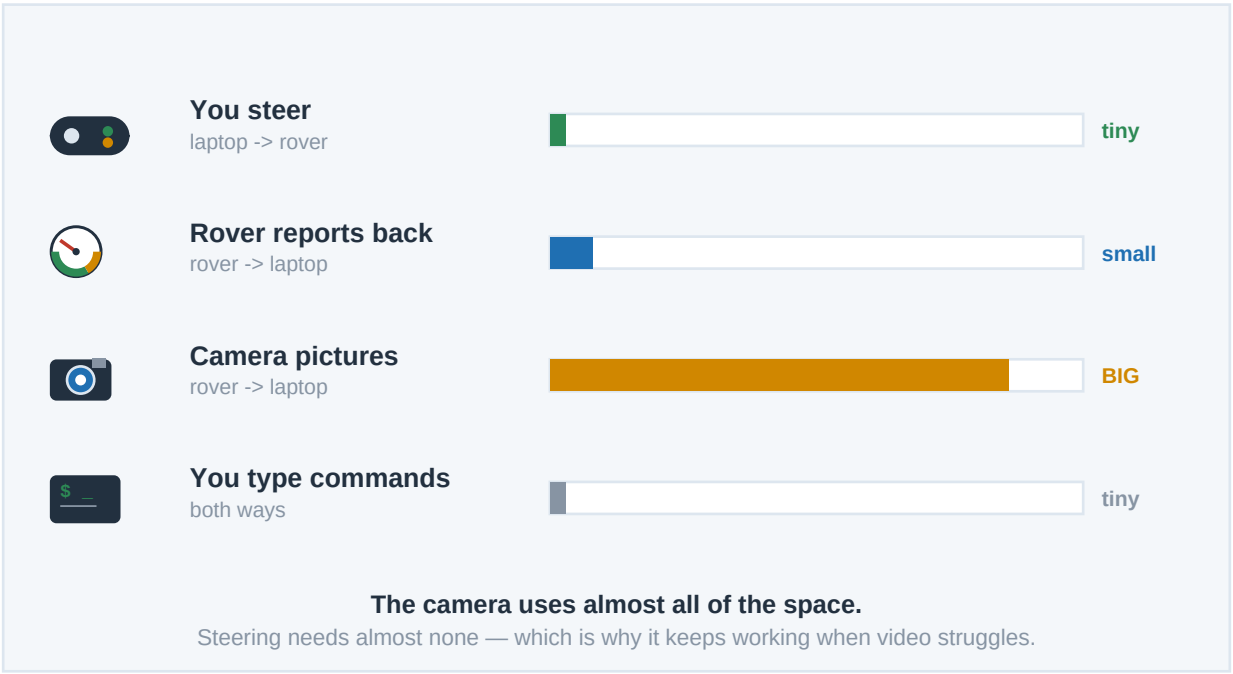
You sit at a **laptop**. The laptop is joined by a **cable** to a **Wi-Fi router** standing in the arena. The router talks by radio to the **rover**. Inside the rover a small **ESP32 board** turns the messages into motor movement.

There is only **one wireless step** in the whole chain — router to rover. Everything else is a wire. Fewer radios means fewer things that can go wrong.

*Why the laptop is wired and not on Wi-Fi: the competition rules forbid the laptop from joining the rover's wireless network. Breaking that rule disqualifies the team, so the laptop's Wi-Fi is switched off.*

## What travels over the link

Four different things share that single radio hop. They are very different sizes:




### This is the useful part to remember

Video is about **nine tenths** of everything being sent. Steering is a rounding error next to it.

So when the signal gets weak, the picture goes blocky **first**, and steering keeps working **longest**. If the video looks bad, that is a warning, not an emergency — you can still drive.

## Starting everything, in order

Five steps. Do them in this order — each one needs the one before it.

### 1 Switch the rover on

It takes about a minute to wake up.

### 2 Check they can see each other

On the laptop, ask for the list of rover topics. If the list is empty, stop and fix that first — nothing else will work.

### 3 Start the joystick

The joystick plugs into the **laptop**, not the rover.

### 4 Open Mission Control

The two badges at the top should turn green: **ROVER LINK** and **JETSON**.

### 5 Press START STREAM

The camera pictures appear. Now you are ready to drive.

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### If the rover will not move

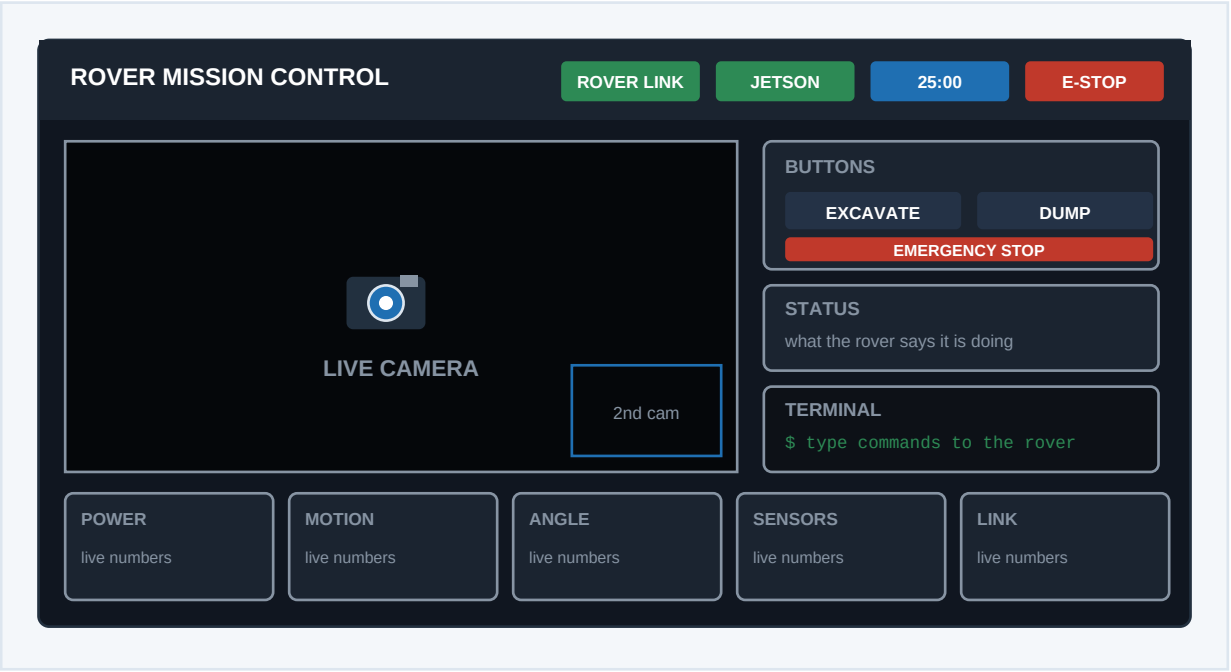
Work through it in this order, because the earlier causes are far more common than the later ones:

**1.** Are both machines using the same radio settings? **2.** Is the joystick actually sending? **3.** Is the rover's controller board plugged in?

*The rover stops itself if it stops hearing from the laptop for half a second. That is deliberate — a rover that keeps driving when it has gone deaf is dangerous.*

## What you see on screen

One window. Camera on the left, controls on the right, numbers along the bottom.




### Reading it at a glance

**Top bar** — two green badges mean the rover is talking to you. The big number is the run clock, counting down from **25 minutes**.

**Left** — what the rover can see. The small inset is the second camera.

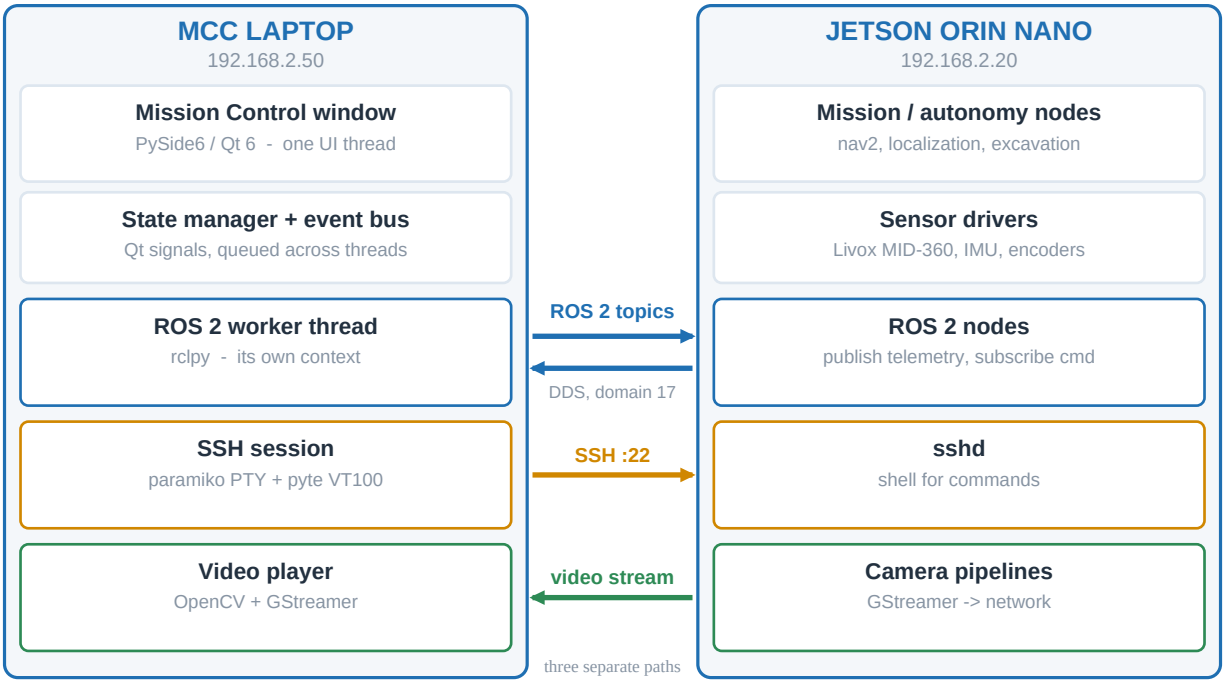
**Right** — the buttons, what the rover says it is doing, and a terminal for typing commands straight to it.

**Bottom** — battery, speed, tilt, sensor health, and link quality.

*One rule worth knowing: the screen shows what the **rover reports**, never what you pressed. If you press a button and the screen does not change, the rover did not do it. That is the display being honest with you, not a bug.*

## What the software is made of

The same two machines as before, but now showing what runs inside each one and which piece talks to which.




### Three paths, kept apart on purpose

**Telemetry** (blue) is the only two-way path: the rover publishes its numbers, the laptop publishes steering and stop commands. Both ends must agree on the same **domain number** and the same DDS software, or they see nothing of each other at all.

**Commands** (amber) is an ordinary SSH login, the same one you would get from a terminal window — it just happens to be drawn inside the app.

**Video** (green) does not use ROS at all. The pictures go straight from the Jetson's cameras to the player.

*Kept apart deliberately: if the video dies, steering still works; if the terminal hangs, telemetry keeps arriving. Each runs on its own thread, so none of them can freeze the window.*

### The pieces, by name

| Where | What it is | Doing what |
|---|---|---|
| Laptop | Ubuntu 22.04 + ROS 2 Humble | The base both machines share |
| Laptop | PySide6 6.11 (Qt 6) | Every window, button and card |
| Laptop | rclpy | Speaks ROS 2 on a background thread |
| Laptop | paramiko 2.9 + pyte 0.8 | The in-app Jetson terminal |
| Laptop | OpenCV 4.5 + GStreamer | Decodes and draws the camera feeds |
| Jetson | JetPack + ROS 2 Humble | Runs the rover's own nodes |
| Jetson | GStreamer | Encodes the cameras and sends them out |
| Jetson | ESP32 firmware | Turns messages into motor movement |